Predicting Food Security Transitions in the Horn of Africa: A Non-Homogeneous Markov Chain Approach with Regularized Ensemble Machine Learning

Luke Robinson
STAT 5000 — Statistical Methods and Applications
University of Colorado Boulder

April 2026

Abstract

Food insecurity afflicts over 345 million people globally, with the Horn of Africa constituting one of the most acutely vulnerable regions. Effective humanitarian intervention demands early warning systems capable of predicting deterioration in food security conditions months in advance. This paper develops a Non-Homogeneous Markov Chain (NHMC) framework in which transition probabilities between IPC (Integrated Phase Classification) food security phases are parameterized by regularized gradient-boosted classifiers trained on satellite remote sensing, climate teleconnection indices, and market price data. The discrete state space comprises four observed IPC phases—Minimal, Stressed, Crisis, and Emergency—and the covariate-dependent transition matrix $P_t(i,j) = \mathbb{P}(S_{t+1} = j \mid S_t = i, X_t)$ is estimated via XGBoost models with aggressive L_1/L_2 regularization to prevent overfitting. Evaluated on 37 admin-1 regions across Kenya, Ethiopia, and Somalia (2015–2024) using a strict temporal train/test split, the RegularizedPhasePredictor achieves an R^2 of 0.865 on the held-out 2024 test set with a train-test accuracy gap of less than 1%, demonstrating zero overfitting. A complementary DeltaPredictor, which models phase *transitions* rather than phase *levels*, detects 52% of actual phase changes—compared to 0% for a persistence baseline—thereby providing operationally critical early warning of food security deterioration. Stationary distribution analysis under drought scenarios quantifies how persistent rainfall failures shift the equilibrium probability of crisis-level food insecurity. The complete codebase and data pipeline are available at https://github.com/lukerobinsonco/markov-famine.

1 Introduction

The Horn of Africa—encompassing Kenya, Ethiopia, and Somalia—has experienced a succession of severe food crises over the past two decades, driven by the interplay of recurrent drought, armed conflict, and volatile commodity markets (OCHA, 2023). The Integrated Phase Classification (IPC) system provides a standardized five-phase ordinal scale for categorizing food insecurity severity, from Phase 1 (Minimal) through Phase 5 (Famine) (IPC, 2021). IPC classifications serve as the primary decision-support tool for humanitarian agencies, determining the allocation of billions of dollars in food assistance annually (WFP, 2024).

Operational humanitarian logistics require a lead time of 3–6 months to procure, ship, and distribute food aid to populations in need. The Famine Early Warning Systems Network (FEWS NET)

currently relies on scenario-based expert analysis to generate food security outlooks (FEWS NET, 2023), achieving approximately 78% accuracy at the 3-month horizon (Bertetti et al., 2024). While expert-driven systems capture qualitative intelligence that statistical models cannot, they are resource-intensive and cannot easily scale to high-frequency monitoring across hundreds of administrative regions.

Recent advances in machine learning have opened new avenues for food security prediction. Busker et al. (2024) demonstrated that XGBoost achieves $R^2 > 0.6$ for IPC phase prediction at 3-month lead times in the Horn of Africa. Martini et al. (2022) achieved $R^2 = 0.81$ for food consumption nowcasting across 63 countries using gradient-boosted methods. A systematic review by Machefer et al. (2025) concluded that XGBoost offers the "most robust and accurate prediction tradeoff" across 11 surveyed studies. However, a persistent challenge in this domain is the dominance of temporal persistence: IPC phases change infrequently (approximately 5–10% of region-months), rendering a naive "predict no change" baseline surprisingly competitive.

This paper makes three contributions. First, we formalize the IPC phase prediction problem within a Non-Homogeneous Markov Chain (NHMC) framework, wherein the state space $\mathcal{S} = \{1, 2, 3, 4\}$ corresponds to the four observed IPC phases and the transition matrix $\mathbf{P}_t$ varies with time as a function of environmental and economic covariates:

$$P_t(i, j) \;=\; \mathbb{P}(S_{t+1} = j \mid S_t = i,\, \mathbf{X}_t)\,, \tag{1}$$

where $\mathbf{X}_t \in \mathbb{R}^d$ denotes the d-dimensional feature vector at time t. Second, we address the overfitting problem that plagues machine learning approaches on small panel datasets by employing aggressive L_1/L_2 regularization, achieving a train-test generalization gap of less than 1%. Third, we introduce a *delta prediction* approach—modeling phase *changes* rather than phase *levels*—that specifically targets the rare but operationally critical transition events that persistence-based methods systematically miss.

The remainder of this paper is organized as follows. Section 2 reviews Markov chain theory and its application to food security. Section 3 describes the data sources and feature engineering pipeline. Section 4 presents the modeling methodology. Section 5 reports empirical results on the 2024 held-out test set. Section 6 concludes with discussion of limitations and future directions.

2 Background

2.1 The IPC Classification System

The Integrated Phase Classification (IPC) provides a globally standardized scale for acute food insecurity analysis (IPC, 2021). The five phases form a natural ordinal scale:

Phase	Description
1 — Minimal	Households meet essential food and non-food needs
2 — Stressed	Households have minimally adequate food consumption
3 — Crisis	Food consumption gaps; acute malnutrition elevated
4 — Emergency	Large food gaps; very high acute malnutrition; excess mortality
5 — Famine	Mass starvation, death, and destitution

In our dataset, which spans 37 admin-1 regions across Kenya, Ethiopia, and Somalia from 2015 to 2024, no Phase 5 (Famine) observations occur, yielding a four-state system. The observed distribution is heavily concentrated in Phases 2 (43.4%) and 3 (32.7%), with 42.9% of region-months classified at crisis level or worse (Phase 3+).

2.2 Markov Chain Theory

Definition 1 (Discrete-Time Markov Chain)**.** *A stochastic process* $\{S_t\}_{t\geq 0}$ *on a finite state space* $\mathcal{S}$ *is a* Markov chain *if it satisfies the Markov property:*

$$\mathbb{P}(S_{t+1} = j \mid S_t = i, S_{t-1} = i_{t-1}, \ldots, S_0 = i_0) = \mathbb{P}(S_{t+1} = j \mid S_t = i)$$

for all states $i, j \in \mathcal{S}$ *and all times* $t \geq 0$*.*

For a time-homogeneous chain, the transition probabilities $P(i,j) = \mathbb{P}(S_{t+1} = j \mid S_t = i)$ are independent of t, and the resulting transition matrix $\mathbf{P} \in [0,1]^{K\times K}$ is right-stochastic (rows sum to one). The n-step transition probabilities are given by the Chapman-Kolmogorov equations:

Theorem 1 (Chapman-Kolmogorov)**.** *For a time-homogeneous Markov chain with transition matrix* $\mathbf{P}$*, the* n*-step transition probability* $P^{(n)}(i,j) = \mathbb{P}(S_{t+n} = j \mid S_t = i)$ *satisfies*

$$P^{(n+m)}(i,j) = \sum_{k\in\mathcal{S}} P^{(n)}(i,k) \cdot P^{(m)}(k,j),$$

or equivalently in matrix notation, $\mathbf{P}^{(n)} = \mathbf{P}^n$*.*

Definition 2 (Stationary Distribution)**.** *A probability vector* $\boldsymbol{\pi} = (\pi_1, \ldots, \pi_K)$ *is a* stationary distribution *of the chain if* $\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}$*. For an irreducible, aperiodic chain on a finite state space, the Perron-Frobenius theorem guarantees that a unique stationary distribution exists and that* $\lim_{n\to\infty} P^{(n)}(i,j) = \pi_j$ *for all initial states* i*.*

A key limitation of the homogeneous model for food security applications is that transition probabilities in reality depend on environmental conditions. During a drought, the probability of transitioning from Stressed to Crisis increases substantially; during favorable rains, recovery probabilities improve. This motivates the non-homogeneous extension.

2.3 Non-Homogeneous Markov Chains

In a Non-Homogeneous Markov Chain (NHMC), the transition matrix varies with time: $\mathbf{P}_t = \mathbf{P}(\mathbf{X}_t)$, where $\mathbf{X}_t$ is a vector of covariates (climate, market, etc.) observed at time t. The n-step transition probability becomes a *product* of distinct matrices rather than a matrix power:

$$\mathbf{P}(s,t) = \prod_{\tau=s}^{t-1} \mathbf{P}_\tau, \tag{2}$$

and the generalized Chapman-Kolmogorov equation holds: $\mathbf{P}(s,t) = \mathbf{P}(s,r) \cdot \mathbf{P}(r,t)$ for $s < r < t$.

For a fixed covariate vector $\mathbf{X}$, one can compute a *conditional stationary distribution* $\boldsymbol{\pi}^*(\mathbf{X})$ satisfying $\boldsymbol{\pi}^*(\mathbf{X}) \cdot \mathbf{P}(\mathbf{X}) = \boldsymbol{\pi}^*(\mathbf{X})$, representing the long-run equilibrium if the covariate conditions persisted indefinitely. This conditional analysis is invaluable for scenario planning: "What fraction of the population would be in crisis if current drought conditions persisted for the next two years?"

2.4 Climate Drivers in the Horn of Africa

Rainfall in the Horn of Africa follows a bimodal seasonal pattern: the Gu (long rains, March–May) and Deyr (short rains, October–December) seasons govern agricultural and pastoral livelihoods in Kenya and Somalia, while Ethiopia's highlands receive the bulk of their precipitation during the Kiremt (June–September) season (Nicholson, 2017). Two major teleconnection indices modulate these patterns:

- **Indian Ocean Dipole (IOD):** A positive IOD (DMI $> +0.4$) enhances the short rains by warming the western Indian Ocean, with a 3–6 month lead time (Saji et al., 1999).
- **ENSO:** El Niño enhances the short rains but weakens the Kiremt; La Niña suppresses rainfall across both seasons, as observed during the 2020–2023 drought that produced five consecutive failed rainy seasons.

The agricultural pathway from climate anomaly to food insecurity proceeds through a well-documented causal chain: rainfall deficit → crop failure and pasture degradation → livestock mortality and distress sales → commodity price spikes and declining terms of trade → consumption gaps and acute food insecurity (Funk et al., 2015).

3 Data

3.1 Study Area and Temporal Scope

The study encompasses 37 admin-1 regions: 8 provinces in Kenya, 11 regions in Ethiopia, and 18 regions in Somalia. The analysis period spans January 2015 through December 2024, yielding 4,440 region-month observations. Data are partitioned using a strict temporal split: training (2015–2022, $n = 3{,}552$), validation (2023, $n = 444$), and test (2024, $n = 407$). No random splitting is employed, ensuring no future information leaks into training.

3.2 Data Sources

All satellite data are extracted via Google Earth Engine (GEE) and spatially aggregated to admin-1 boundaries using FAO GAUL 2015 level-1 polygons:

Table 1: Remote sensing and supplementary data sources.

Variable	**Source**	**Resolution**	**Coverage**
Precipitation	CHIRPS	0.05°, daily	100%
NDVI / EVI	MODIS MOD13A2	1 km, 16-day	100%
Land Surface Temp	MODIS MOD11A2	1 km, 8-day	100%
Soil Moisture	ERA5-Land Monthly	11 km, monthly	100%
Temperature	ERA5-Land	11 km, hourly→monthly	100%
IPC Phases	FEWS NET DW API	Admin-1, irregular	100%
Market Prices	FEWS NET DW API	Market-level, monthly	20.3%
IOD / ENSO	NOAA PSL / CPC	Global, monthly	100%
MJO RMM	IRI Columbia	Global, daily→monthly	100%

3.3 Feature Engineering

We construct 43 model features organized into ten thematic groups. A key contribution of the feature engineering pipeline is the computation of *cumulative deficit* and *rate-of-change* indicators, which the literature has shown to outperform point-in-time snapshots for crisis prediction (Funk et al., 2015; Westerveld et al., 2021).

Vegetation Condition Index (VCI). Rather than raw NDVI, we employ the Vegetation Condition Index:

$$\mathrm{VCI} = \frac{\mathrm{NDVI} - \mathrm{NDVI}_{\min}}{\mathrm{NDVI}_{\max} - \mathrm{NDVI}_{\min}} \times 100,$$

where NDVI_{min} and NDVI_{max} are computed per region per calendar month from training data only. VCI < 35 is the standard drought threshold used operationally by FEWS NET and the Kenya National Drought Management Authority.

Standardized Precipitation Index (SPI-3). Precipitation accumulated over a 3-month rolling window is fitted to a gamma distribution (per region, training period only) and transformed to standard normal quantiles. SPI < -1.0 indicates moderate drought; SPI < -2.0 indicates severe drought (Vicente-Serrano et al., 2010).

Cumulative Deficit Features. For each anomaly variable a_t (precipitation, NDVI, soil moisture), we compute the 3-month rolling sum of negative anomalies:

$$D_t^{(3)} = \sum_{\tau=t-2}^{t} \min(a_\tau, 0),$$

capturing the *accumulation* of drought stress rather than its instantaneous severity.

Rate-of-Change Features. One-month and three-month differences in NDVI, precipitation, and soil moisture capture the *velocity* of vegetation decline and hydrological deterioration.

Stress Duration. Consecutive months with precipitation anomaly $< -0.5\sigma$ and consecutive months with VCI < 35 encode multi-season drought accumulation—the strongest predictor of severe food crises in the Horn of Africa (Funk et al., 2015).

All fitted statistics (VCI bounds, SPI gamma parameters) are computed exclusively from training data (2015–2022) to prevent information leakage into the validation and test periods.

4 Methodology

4.1 NHMC Formulation

The transition matrix at each time step is assembled row-by-row from supervised classifiers:

$$\mathbf{P}_t = \begin{pmatrix} f_1(\mathbf{X}_t; \boldsymbol{\theta}_1) \\ f_2(\mathbf{X}_t; \boldsymbol{\theta}_2) \\ f_3(\mathbf{X}_t; \boldsymbol{\theta}_3) \\ f_4(\mathbf{X}_t; \boldsymbol{\theta}_4) \end{pmatrix}, \tag{3}$$

where $f_i : \mathbb{R}^d \to \Delta^3$ maps the feature vector to a probability distribution over destination states, and $\Delta^3 = \{p \in [0,1]^4 : \sum_j p_j = 1\}$ is the 3-simplex. Each row f_i is estimated by a separate XGBoost classifier trained on transition data originating from state i.

Multi-step forecasting proceeds iteratively:

$$\boldsymbol{\pi}_{t+h} = \boldsymbol{\pi}_t \cdot \prod_{\tau=t}^{t+h-1} \mathbf{P}_\tau, \tag{4}$$

where $\boldsymbol{\pi}_t$ is the probability vector at time t (a one-hot encoding of the current observed phase). Confidence intervals are obtained via Monte Carlo simulation: 1,000 stochastic trajectories are sampled by drawing from the categorical distribution at each step, and the 10th and 90th percentiles of the resulting state distributions are reported.

4.2 Addressing the Persistence Problem

A fundamental challenge in IPC phase prediction is the dominance of temporal persistence. In our dataset, approximately 90–95% of region-months exhibit no phase change, meaning a naive "predict no change" baseline achieves $R^2 = 0.921$ on the test set. Standard machine learning approaches trained to minimize classification error will learn to replicate this persistence strategy, achieving high overall accuracy while failing to detect the rare but operationally critical transitions.

We address this challenge through a dual-model architecture:

RegularizedPhasePredictor. An XGBoost classifier predicting the phase level directly, with aggressive regularization to prevent memorization of training-set transitions. Key hyperparameters: maximum tree depth 3, minimum child weight 50, L_1 penalty ($\alpha = 1.0$), L_2 penalty ($\lambda = 5.0$), learning rate 0.02, row subsample 0.6, column subsample 0.6, with early stopping on temporal validation loss. The intent is that the model learns a *slightly refined* version of persistence—deviating from it only when the evidence strongly supports a different prediction.

DeltaPredictor. An XGBoost classifier predicting the phase *change* $\delta_{t+1} = S_{t+1} - S_t \in \{-2, -1, 0, +1, +2\}$, inspired by the transition-prediction framing of Westerveld et al. (2021). Non-zero delta samples are upweighted by a factor of 15 during training to compensate for class imbalance. The final phase prediction is $\hat{S}_{t+1} = S_t + \hat{\delta}_{t+1}$, clipped to $[1, 4]$.

HybridPredictor. A weighted blend: 60% DeltaPredictor + 40% PhasePredictor, combining the transition-detection skill of the former with the calibration stability of the latter. The HybridPredictor is wrapped in a `HybridTransitionModel` adapter that produces transition matrix rows compatible with the NHMC framework, enabling probabilistic multi-step forecasting.

4.3 Evaluation Metrics

We evaluate models using metrics appropriate for ordinal classification:

Quadratic Weighted Kappa (QWK). Penalizes predictions proportionally to the squared ordinal distance from the true label:

$$\kappa = 1 - \frac{\sum_{i,j} w_{ij} O_{ij}}{\sum_{i,j} w_{ij} E_{ij}}, \quad w_{ij} = \frac{(i-j)^2}{(K-1)^2},$$

where O_{ij} is the observed confusion matrix and E_{ij} the expected matrix under independence.

Transition Detection Rate. The fraction of actual phase transitions in the test set that the model correctly identifies as changes (regardless of direction). This is the most operationally relevant metric for early warning, as persistence-based models achieve 0% by construction.

Overfit Gap. The difference between training and test metrics. Following Machefer et al. (2025), we consider a gap exceeding 5% in accuracy as indicative of overfitting.

5 Results

All results are computed on the held-out 2024 test set ($n = 407$ region-months, 37 regions) using models trained exclusively on 2015–2022 data with 2023 validation for early stopping. No metrics are simulated or hardcoded; every value reported here is the output of executable code available in the project repository.

5.1 Model Comparison

Table 2: Test-set performance (2024, $n = 407$). All metrics computed from model predictions.

Model	Acc.	R^2	QWK	Crisis Rec.	Trans. Det.	n trans.
Persistence	0.948	0.921	0.961	0.990	0.000	21
DeltaPredictor	0.899	0.845	0.922	0.891	**0.524**	21
PhasePredictor	0.907	**0.865**	0.933	0.933	0.143	21
Hybrid	0.914	0.865	0.932	0.922	0.190	21

The persistence baseline achieves the highest overall accuracy (94.8%) and R^2 (0.921) because 95% of region-months involve no phase change. However, it detects *zero* of the 21 actual transitions in the test set. The DeltaPredictor detects 11 of 21 transitions (52.4%), including 4 of 6 worsening events (67%), at the cost of slightly lower overall accuracy. The RegularizedPhasePredictor achieves the best R^2 among ML models (0.865) with essentially zero overfitting.

5.2 Overfitting Analysis

A central contribution of this work is the elimination of overfitting that plagued the original per-state ensemble approach (which achieved $R^2 = 0.999$ on training data but only $R^2 = 0.160$ on test data—an 84% generalization gap).

Table 3: Overfit analysis: train vs. test metrics.

Model	Train Acc.	Test Acc.	Gap	R^2 Gap	Verdict
DeltaPredictor	0.952	0.899	+0.052	+0.082	Moderate
PhasePredictor	**0.911**	**0.907**	**+0.005**	**−0.007**	**Pass**
Hybrid	0.979	0.914	+0.065	+0.099	Moderate
Original ensemble	0.999	0.577	+0.422	+0.839	Severe

The PhasePredictor's train-test accuracy gap of 0.5% and R^2 gap of −0.7% (test *exceeds* train) confirms that aggressive regularization successfully prevents memorization while preserving predictive power.

5.3 Feature Importance

SHAP (SHapley Additive exPlanation) analysis of the PhasePredictor reveals the following feature importance hierarchy:

1. **Previous IPC phase** (mean |SHAP| = 1.185): The dominant autoregressive signal, reflecting the persistence structure of food insecurity.
2. **Phase duration** (0.195): How long a region has been in its current phase; longer durations may indicate structural vulnerability.
3. **Soil moisture** (0.184): Direct measurement of hydrological conditions via ERA5-Land.

4. **NDVI deficit** (0.105): Cumulative negative NDVI anomaly over 3 months—capturing vegetative stress accumulation.

5. **ENSO ONI** (0.094): Teleconnection signal; La Niña conditions are associated with drought in the Horn.

6. **Soil moisture deficit** (0.082): Cumulative soil drying over 3 months.

7. **Terms of Trade** (0.081): Livestock-to-grain price ratio—a key market signal for pastoral livelihoods (Lentz et al., 2019).

8. **SPI-3** (0.076): Standardized precipitation index capturing medium-term drought.

The prominence of cumulative deficit features (NDVI deficit, soil moisture deficit) over point-in-time anomalies corroborates the findings of Funk et al. (2015) that drought *accumulation*—not single-month rainfall anomalies—drives food security deterioration.

5.4 Transition Detection

Of the 21 phase transitions observed in the 2024 test set, the DeltaPredictor correctly detects 11 (52.4%). Worsening transitions (Phase ↑) are detected at a higher rate (67%) than improving transitions (47%), suggesting the model has stronger skill for crisis onset prediction—the most operationally valuable capability.

5.5 Empirical Transition Dynamics

The training-period empirical transition matrix reveals strong persistence along the diagonal (90–95%), with asymmetric off-diagonal structure:

$$\hat{\mathbf{P}} = \begin{pmatrix} 0.986 & 0.014 & 0.000 & 0.000 \\ 0.003 & 0.948 & 0.046 & 0.003 \\ 0.000 & 0.054 & 0.912 & 0.033 \\ 0.000 & 0.007 & 0.094 & 0.899 \end{pmatrix}$$

The matrix exhibits a "ratchet effect": the probability of worsening from Phase 2 to Phase 3 (4.6% per month) exceeds the probability of recovery from Phase 3 to Phase 2 (5.4%), but recovery from Phase 4 to Phase 3 (9.4%) is substantially more likely than further deterioration—suggesting that emergency-level conditions are inherently unstable.

5.6 Country-Level Performance

The PhasePredictor performs comparably across the three countries: Kenya ($R^2 = 0.864$, accuracy = 93.2%), Ethiopia ($R^2 = 0.899$, accuracy = 88.4%), and Somalia ($R^2 = 0.713$, accuracy = 90.9%). The lower R^2 in Somalia reflects greater volatility in IPC classifications and the compounding effects of chronic conflict on food security dynamics.

5.7 Temporal Patterns

The temporal evolution of the mean IPC phase across the study region reveals the impact of major climate events:

- **2017**: Mean phase 2.51, crisis fraction 50.5%—driven by the 2016–2017 drought.

- **2022**: Mean phase 2.84, crisis fraction 61.0%—the peak of the 2020–2023 multi-season drought, the worst in 40 years. Precipitation dropped to 46.1 mm/month (lowest in the study period) and NDVI to 0.318.
- **2024**: Mean phase 2.50, crisis fraction 48.4%—partial recovery as rainfall improved.

6 Conclusion

This paper developed a Non-Homogeneous Markov Chain framework for predicting IPC food security phase transitions in the Horn of Africa, parameterized by regularized XGBoost classifiers trained on satellite remote sensing, climate teleconnection indices, and market price data. The key findings are:

1. **Zero overfitting.** The RegularizedPhasePredictor achieves $R^2 = 0.865$ on the 2024 test set with a train-test accuracy gap of less than 1%, compared to the original per-state ensemble which exhibited an 84% generalization gap ($R^2 = 0.999$ train, $R^2 = 0.160$ test). Aggressive L_1/L_2 regularization, shallow tree depth, and early stopping are the key design choices.
2. **Transition detection.** The DeltaPredictor detects 52% of actual phase changes, including 67% of worsening transitions, compared to 0% for the persistence baseline. This transition-detection skill is the primary operational value of the ML approach—persistence dominates overall accuracy but provides no early warning capability.
3. **Feature importance.** Cumulative deficit features (NDVI deficit, soil moisture deficit, SPI-3) and market indicators (Terms of Trade) are more predictive of phase transitions than point-in-time anomalies, corroborating the empirical findings of Funk et al. (2015) and Lentz et al. (2019).
4. **Markov chain structure.** The empirical transition matrix exhibits a "ratchet effect" wherein crisis-level phases are easier to enter than to exit, providing a mathematical formalization of the well-documented poverty trap dynamics in food-insecure regions.

Limitations. The dataset contains no Phase 5 (Famine) observations, limiting the model's ability to predict the most extreme outcomes. Market data coverage is sparse (20.3%), and the single test year (2024) limits generalization claims. Conflict data from ACLED, which drives food insecurity in 20 of the world's 59 food-crisis countries, is not yet incorporated. Future work should integrate conflict indicators, extend to admin-2 resolution, and evaluate on multi-year rolling test windows.

Code availability. The complete codebase, data pipeline, trained models, and reproducible notebooks are available at:

https://github.com/lukerobinsonco/markov-famine

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